

Syntax reference

Octave and Python (NumPy) — MATH 107

Last updated: September 20, 2026.

Pick **one** language for the term. Lab hubs show both; this page is the growing cheat sheet. Full docs: Lachniet PDF and NumPy links on [Resources](#).

Indexing reminder: Boyd's x_1 is the first entry. Octave is **1-based** ($\mathbf{x}(1)$). Python / NumPy is **0-based** ($\mathbf{x}[0]$).

Vectors and basics

Task	Octave	Python (NumPy)
Column vector	<code>x = [1; 2; 3]</code>	<code>x = np.array([1.0, 2.0, 3.0])</code>
Length n	<code>length(x)</code>	<code>len(x)</code> or <code>x.size</code>
First / second entry	<code>x(1), x(2)</code>	<code>x[0], x[1]</code>
Add / scale	<code>x+y, 2*x</code>	<code>x+y, 2*x</code>
Dot product $x^T y$	<code>x'*y</code>	<code>x @ y</code> or <code>np.dot(x,y)</code>
Plot in 2D	<code>plot(x(1), x(2), 'o')</code>	<code>plt.scatter(x[0], x[1])</code> (after import <code>matplotlib.pyplot</code> as <code>plt</code>)

Norm, distance, angle

Task	Octave	Python (NumPy)
Norm $\ x\ $	<code>norm(x)</code>	<code>np.linalg.norm(x)</code>
Distance $\ x - y\ $	<code>norm(x-y)</code>	<code>np.linalg.norm(x-y)</code>
Cosine similarity	<code>(a'*b)/(norm(a)*norm(b))</code>	<code>(a @ b)/(np.linalg.norm(a)*np.linalg.norm(b))</code>
Angle (degrees)	<code>acosd(cos_th)</code>	<code>np.degrees(np.arccos(np.clip(cos_th,-1,1)))</code>

Many candidates (rows of a matrix)

Store each candidate as a **row** of **Z**. Query **x** is a column in Octave examples below (transpose with **x'** when subtracting from rows).

Task	Octave	Python (NumPy)
Stack candidates as rows	<code>Z = [1 2; 3 4; 5 6]</code>	<code>Z = np.array([[1,2],[3,4],[5,6]])</code>
All distances to x	<code>d = sqrt(sum((Z - x').^2, 2))</code>	<code>d = np.linalg.norm(Z - x, axis=1)</code>
Nearest index	<code>[~, j] = min(d) → 1-based</code>	<code>j = np.argmin(d) → 0-based</code>
Smallest distance	<code>min(d)</code>	<code>np.min(d)</code>

Unit I labs using this sheet

Lab	Focus
Lab 0	Enter vectors, arithmetic, basic plot
Lab 1	Short: distances and nearest neighbor
Lab 2	Software practice: distances, scale, optional k -means assignment

Later labs will add matrix, solve, eigen, and least-squares rows here.